

Week 01 — Stationarity, Autocovariance & Ergodicity

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The big picture. Time series are dependent data; the whole analysis rests on first/second moments that are invariant to time shifts (weak stationarity) and on the autocovariance function.

A **time series** $\{X_t\}$ is both the data recorded in time order *and* the stochastic process $\{X_t : t \in T\}$ on $(\Omega, \mathcal{F}, \mathbb{P})$ of which the data is one realisation. Here $T = \mathbb{Z}$ and the sampling step $\Delta = 1$. Unlike classical statistics, the observations are **dependent**: “there are many more ways to be dependent than to be independent.” We usually see only *one* trajectory, yet must still recover population information from it.

The central challenge of the week

With a single sample of length n , how can we estimate a “population” mean or variance? Answer: if the structure does not change over time (**stationarity**), then a *time average* can stand in for an average over many realisations.

Key definitions

Definition — Weak (second-order) stationarity

$\{X_t\}$ is **weakly stationary** (also: second-order, covariance stationary) if all first- and second-order moments exist, are finite, and are invariant to time shifts. Concretely, for all t, s, τ :

$$(i) \mathbb{E}[X_t] = \mu; \quad (ii) \text{Var}(X_t) = \sigma^2 < \infty; \quad (iii) \mathbb{E}[X_t X_{t+\tau}] = \mathbb{E}[X_s X_{s+\tau}].$$

Condition (iii) forces $\mathbb{E}[X_t X_{t+\tau}]$ to be a function of the lag τ *only*.

Definition — Strict (strong) stationarity

$\{X_t\}$ is **strictly stationary** if, for every n and all $t_1, \dots, t_n$, the joint law of $(X_{t_1}, \dots, X_{t_n})$ equals that of $(X_{t_1+\tau}, \dots, X_{t_n+\tau})$: the *whole* distribution is shift-invariant, not just the first two moments.

Weak vs strict: neither implies the other

Strict $\not\Rightarrow$ **weak**: i.i.d. Student- t on 1 d.o.f. (Cauchy) is strictly stationary but has no finite moments, so (ii) fails. **Weak** $\not\Rightarrow$ **strict**: weak stationarity says nothing about moments of order ≥ 3 . The two *coincide* for a **Gaussian** process (every joint law is multivariate normal, hence fully fixed by mean and covariance).

Definition — Autocovariance (ACVS, γ_τ) and autocorrelation (ACF, ρ_τ)

For weakly stationary $\{X_t\}$,

$$\gamma_\tau = \text{Cov}(X_t, X_{t+\tau}), \quad \rho_\tau = \frac{\gamma_\tau}{\gamma_0} = \text{Corr}(X_t, X_{t+\tau}).$$

Properties: $\gamma_0 = \text{Var}(X_t) \geq 0$, symmetry $\gamma_{-\tau} = \gamma_\tau$, and $\rho_0 = 1, |\rho_\tau| \leq 1$. The ACVS is, for a series, the analogue of the covariance matrix of a vector.

Definition — ACVS estimators

Given $X_1, \dots, X_n$ with sample mean $\bar{X}$, for $|\tau| \leq n-1$ (both 0 otherwise):

$$\tilde{\gamma}_\tau = \frac{1}{n-|\tau|} \sum_{t=1}^{n-|\tau|} (X_t - \bar{X})(X_{t+|\tau|} - \bar{X}), \quad \hat{\gamma}_\tau = \frac{1}{n} \sum_{t=1}^{n-|\tau|} (X_t - \bar{X})(X_{t+|\tau|} - \bar{X}).$$

$\tilde{\gamma}_\tau$ (divisor $n-|\tau|$) is *unbiased* (with known mean) but need not be positive semi-definite. $\hat{\gamma}_\tau$ (divisor n) is *biased* ($\mathbb{E}[\hat{\gamma}_\tau] = \frac{n-|\tau|}{n} \gamma_\tau$) but **is** positive semi-definite: this is the default. Plug-in ACF: $\hat{\rho}_\tau = \hat{\gamma}_\tau / \hat{\gamma}_0$.

Definition — Mean ergodicity

$\{X_t\}$ is **mean ergodic** if its first two moments are finite and $\bar{X} \xrightarrow{P} \mathbb{E}[X_t]$ as $n \rightarrow \infty$: “temporal averages = population averages.” Ergodicity and stationarity are *not* equivalent.

Key results & formulas

Theorem — Existence (Kolmogorov)

A family of finite-dimensional distributions is that of a stochastic process *iff* it is **consistent** (marginalising out one coordinate returns the law of the rest):

$$\lim_{x_i \rightarrow \infty} F_{\mathbf{t}}(\mathbf{x}) = F_{\mathbf{t}(i)}(\mathbf{x}(i)).$$

Why: it guarantees that an infinite-index process $\{X_t\}_{t \in \mathbb{Z}}$ really exists given only its finite marginals — which legitimises modelling the whole time axis even though we observe only n points.

Proposition — The ACVS is positive semi-definite

For any n , all $t_1, \dots, t_n \in \mathbb{Z}$ and all $a_1, \dots, a_n \in \mathbb{R}$:

$$\sum_{j=1}^n \sum_{k=1}^n a_j a_k \gamma_{j-k} \geq 0.$$

Idea: set $W = \sum_j a_j X_j$; then $0 \leq \text{Var}(W) = \sum_{j,k} a_j a_k \gamma_{j-k}$. *Why:* this is the *defining* algebraic property of a genuine autocovariance — a sequence that is not p.s.d. cannot be an ACVS.

Proposition — Variance of $\bar{X}$ and consistency

If $\sum_\tau |\gamma_\tau| < \infty$, then $\bar{X}$ is unbiased and

$$\text{Var}(\bar{X}) = \frac{1}{n} \sum_{\tau=-(n-1)}^{n-1} \left(1 - \frac{|\tau|}{n}\right) \gamma_\tau, \quad n \text{ Var}(\bar{X}) \xrightarrow{n \rightarrow \infty} \sum_{\tau=-\infty}^{\infty} \gamma_\tau.$$

Hence $\text{Var}(\bar{X}) \rightarrow 0$ and $\bar{X} \xrightarrow{P} \mu$. *Idea:* collect the double sum by lag τ , then apply the **Cesàro summability theorem**. *Why:* this is the prototype of “when can a sample average replace a population average?”

Technique — ACVS of a finite white-noise filter

For $X_t = \sum_k \theta_k \varepsilon_{t-k}$ with ε white noise of variance σ^2 (so $\text{Cov}(\varepsilon_s, \varepsilon_u) = \sigma^2 \mathbf{1}_{\{s=u\}}$):

$$\gamma_\tau = \sigma^2 \sum_j \theta_j \theta_{j+|\tau|}.$$

Only the terms whose noise indices coincide survive — this single reflex powers *all* MA covariance computations.

Why dependence changes everything

$n \text{Var}(\bar{X}) \rightarrow \sum_\tau \gamma_\tau$: precision is driven not by n alone but by the *sum of all correlations*. Strong positive correlation (e.g. AR(1) with $\phi \rightarrow 1$) inflates $\text{Var}(\bar{X})$: many observations, but little “independent information.”

Stationary $\neq$ ergodic

Stationarity says the *structure* does not move over time; ergodicity says *one* trajectory suffices to recover the population mean. Typical counterexample: $X_t = Y$ with Y a fixed random variable — perfectly stationary, yet $\bar{X} \rightarrow Y$ (not a constant), so it is *not* ergodic.

Two-tap filter (MA(1))

$X_t = \theta_1 \varepsilon_t + \theta_2 \varepsilon_{t-1}$ gives $\gamma_0 = (\theta_1^2 + \theta_2^2)\sigma^2$, $\gamma_{\pm 1} = \theta_1 \theta_2 \sigma^2$, $\gamma_\tau = 0$ otherwise, hence

$$\rho_1 = \frac{\theta_1 \theta_2}{\theta_1^2 + \theta_2^2}, \quad |\rho_1| \leq \frac{1}{2}$$

(by AM–GM, equality iff $|\theta_1| = |\theta_2|$). **No MA(1) can have $|\rho_1| > \frac{1}{2}$** : a handy sanity check.

Key takeaways

Key takeaways — the week’s reflexes

- **Test weak stationarity** in order, stop at the first failure: (1) is $\mathbb{E}[X_t]$ constant? (2) is $\text{Var}(X_t)$ finite and constant? (3) does $\text{Cov}(X_t, X_{t+\tau})$ depend on τ only? Classic trap: a covariance that keeps a t (e.g. $\cos(ct) \cos(c(t+\tau))$).
- **Strict $\Leftrightarrow$ weak** only for *Gaussian* processes; in general neither implies the other.
- **ACVS**: remember $\gamma_0 = \text{Var} \geq 0$, $\gamma_{-\tau} = \gamma_\tau$, positive semi-definite. A non-p.s.d. sequence is not an ACVS.
- **Estimators**: by default divide by n ($\hat{\gamma}_\tau$, biased but p.s.d.). Unbiasedness is not the goal; low MSE is.
- **Filtered white noise**: $\gamma_\tau = \sigma^2 \sum_j \theta_j \theta_{j+|\tau|}$ — keep only the coinciding noise indices.
- **Consistency of $\bar{X}$** holds under $\sum_\tau |\gamma_\tau| < \infty$; but dependence, not just n , controls precision. Stationary $\neq$ ergodic.